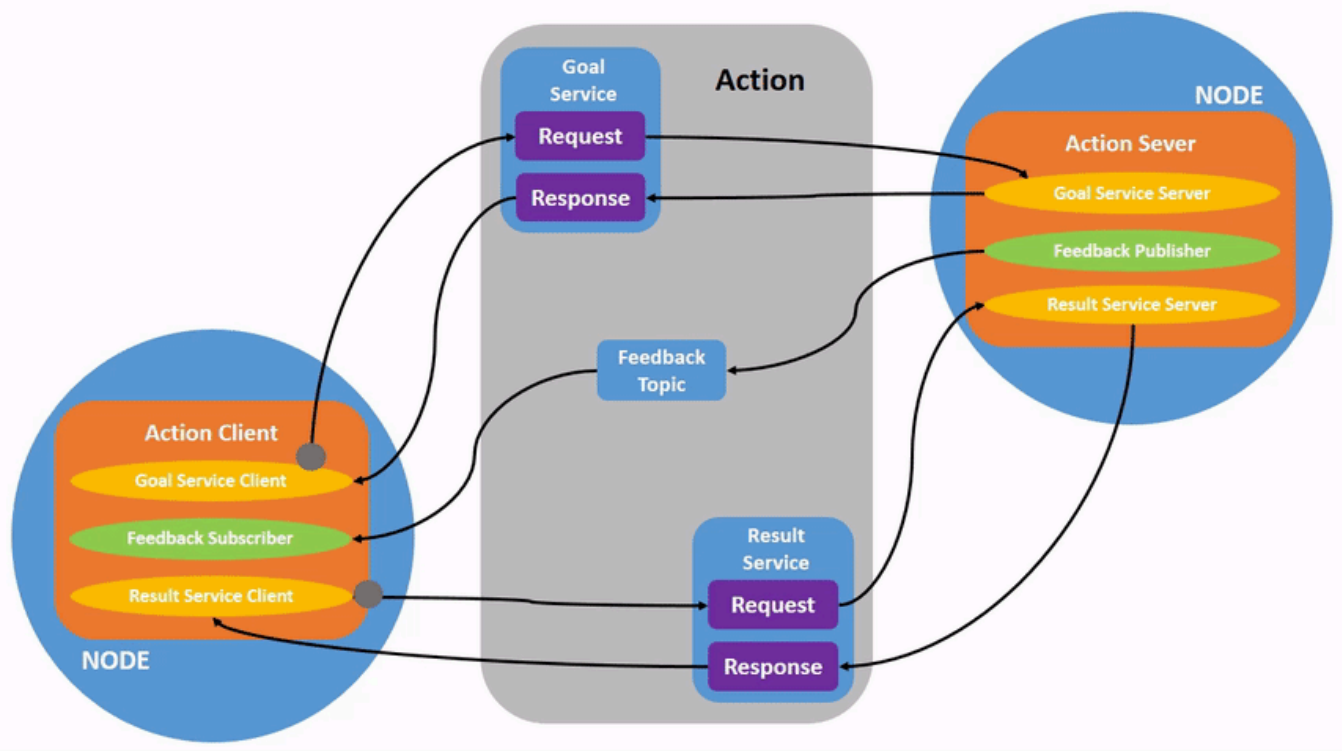
9. ROS2 Action Communication

1. Introduction to Action Communication

Action communication is a communication model with continuous feedback. Between the two parties, the client sends the request data to the server, and the server responds with the result to the client. However, during the process from the server receiving the request to generating the final response, continuous feedback information is sent to the client.

The action communication client/server model is as follows:



2. Case Introduction

The action client submits an integer data N. The action server receives the request data and accumulates all the integers between 1-N, returns the final result to the action client, and each time it accumulates it calculates the current computation progress and feeds it back to the action client.

3. Creating a New Function Package

3.1. Creating the Action Communication Interface Function Package

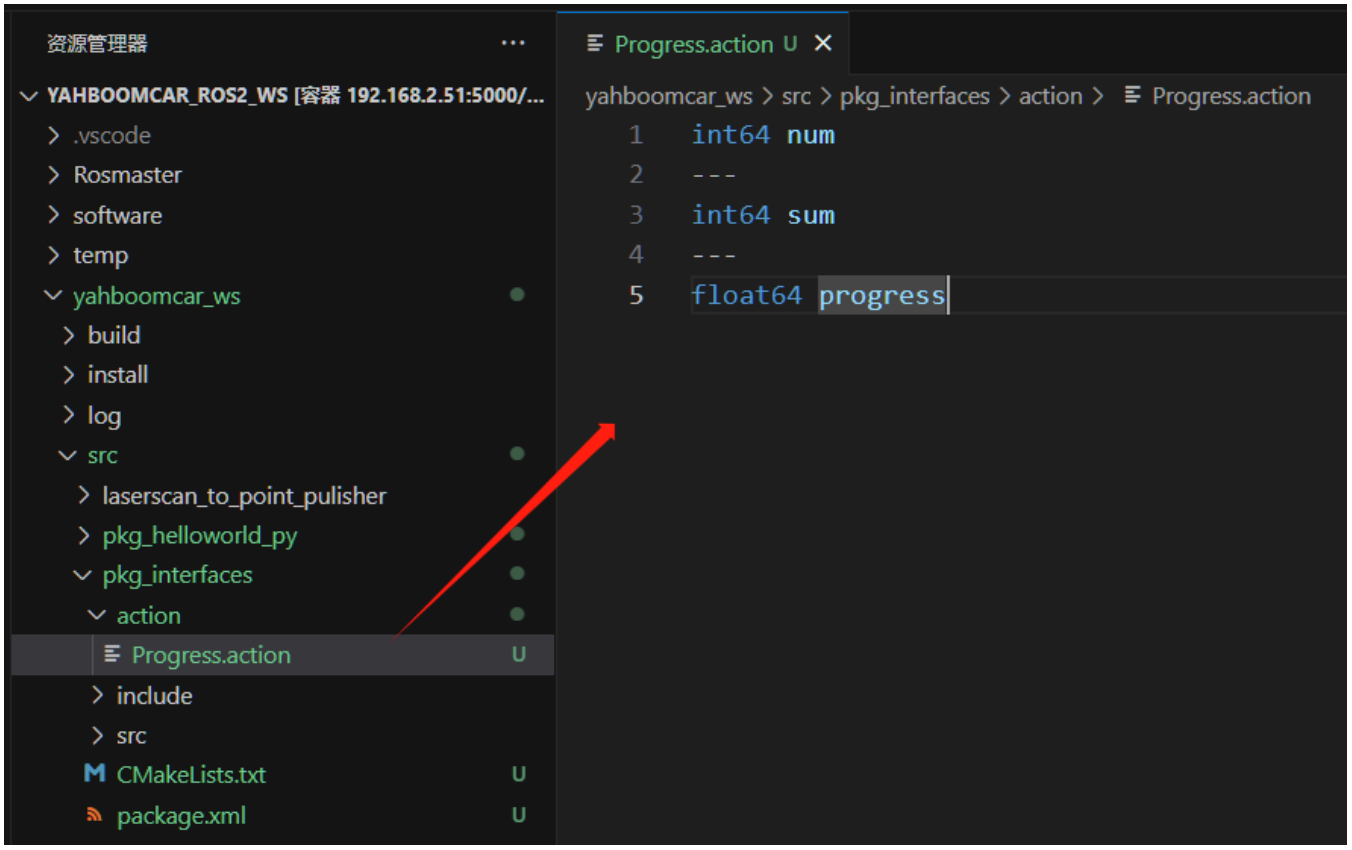
1. Action communication requires creating the action communication interface first

- Create a new pkg_interfaces function package in the src directory of the workspace

```
ros2 pkg create --build-type ament_cmake pkg_interfaces
```

2. Next, create an action folder under the pkg_interfaces function package, and create a new file [Progress.action] inside the action folder. The file content is as follows:

```
int64 num
---
int64 sum
---
float64 progress
```



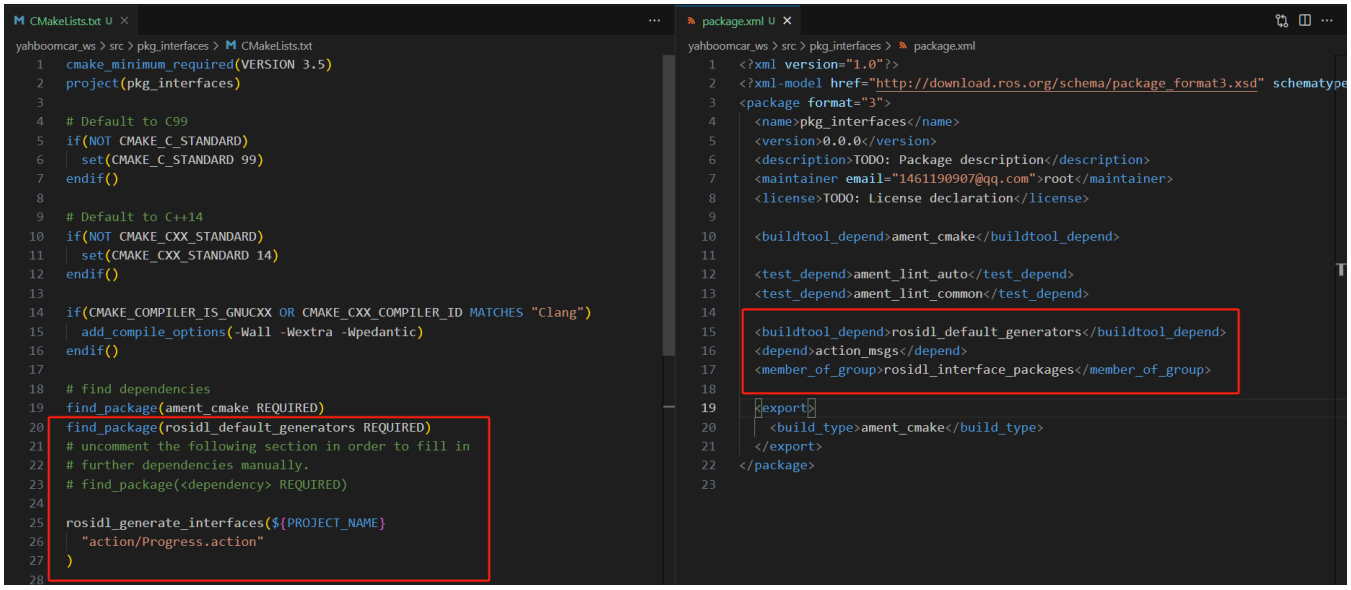
3. Some dependency packages need to be added in package.xml. The specific content is as follows:

```
<buildtool_depend>rosidl_default_generators</buildtool_depend>
<exec_depend>rosidl_default_runtime</exec_depend>
<depend>action_msgs</depend>
<member_of_group>rosidl_interface_packages</member_of_group>
```

4. Add the following configuration in CMakeLists.txt:

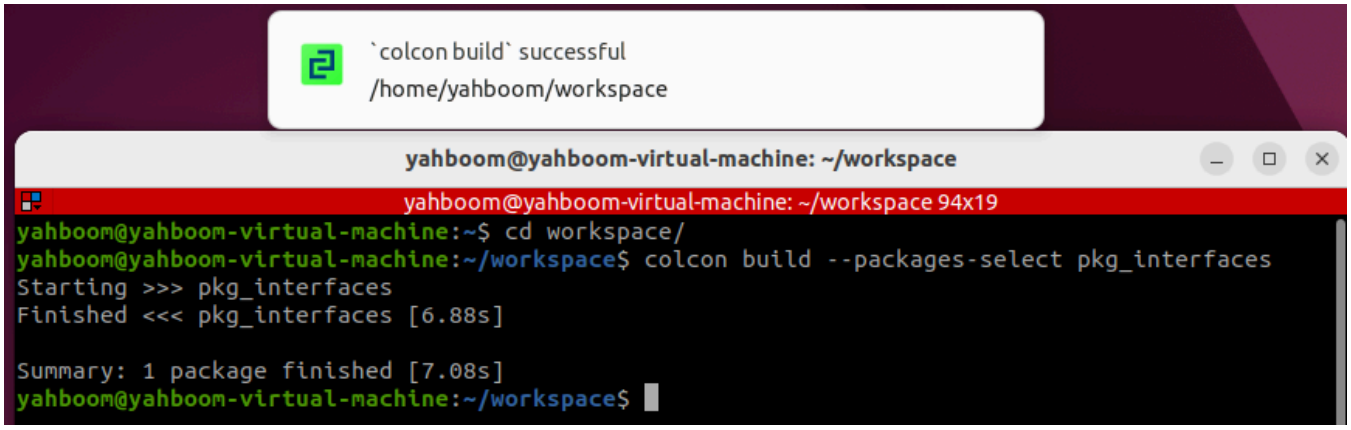
```
find_package(rosidl_default_generators REQUIRED)

rosidl_generate_interfaces(${PROJECT_NAME}
  "action/Progress.action"
)
```



5. Build the function package:

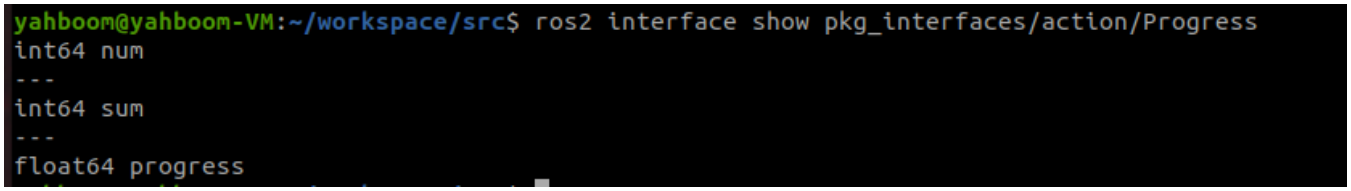
```
colcon build --packages-select pkg_interfaces
```



6. After the build is complete, the C++ and Python files corresponding to the `Progress.action` file will be generated in the install directory under the workspace. We can also enter the workspace in the terminal and check the file definition and whether the build is normal with the following command:

```
source install/setup.bash
ros2 interface show pkg_interfaces/action/Progress
```

Under normal circumstances, the terminal will output the content consistent with the `Progress.action` file

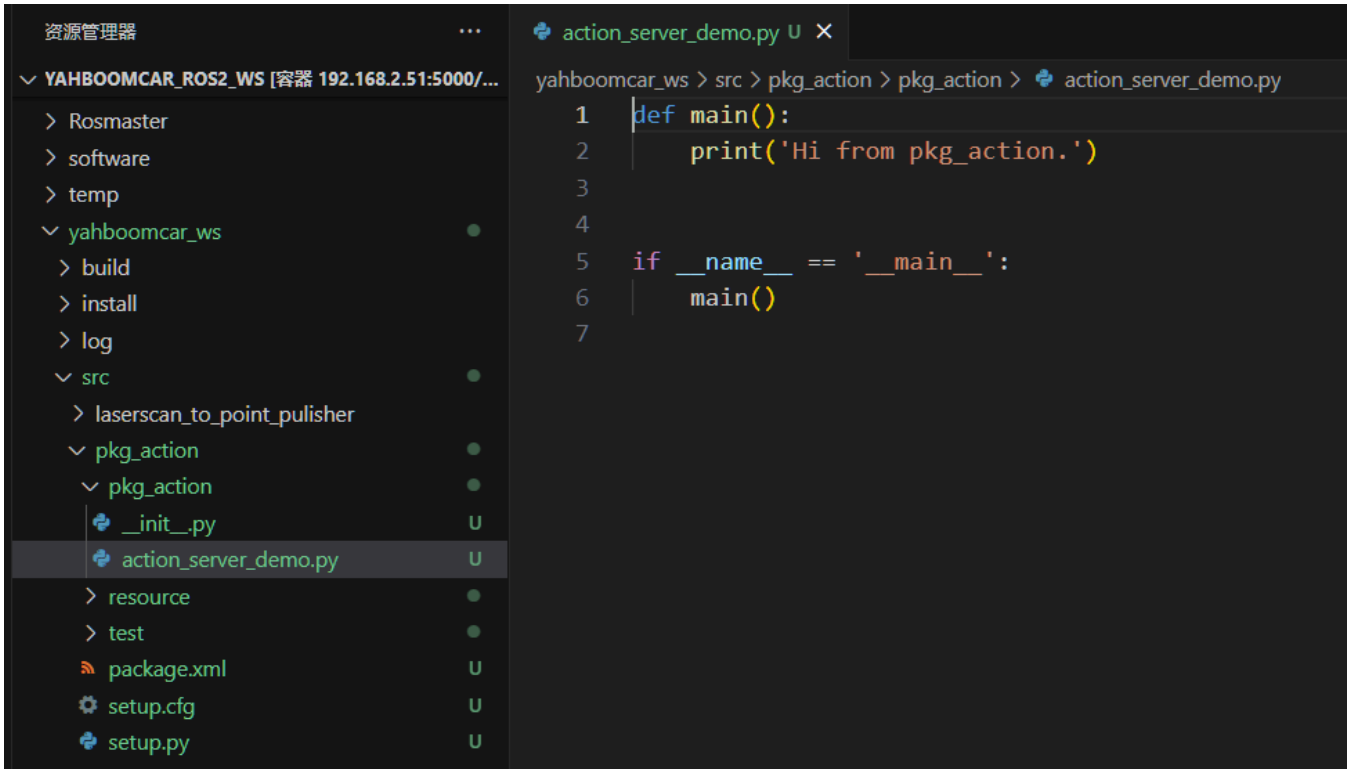


3.2. Creating the Action Communication Function Package

- Create a new pkg_action function package in the src directory of the workspace

```
ros2 pkg create pkg_action --build-type ament_python --dependencies rclpy pkg_interfaces --
node-name action_server_demo
```

After executing the command above, the pkg_action function package is created, an action_server_demo node is created at the same time, and the relevant configuration files have already been set up



4. Server Implementation

4.1 Create the Server

Next, edit [action_server_demo.py] to implement the server function and add the following code:

```
import time
import rclpy
from rclpy.action import ActionServer
from rclpy.node import Node

from pkg_interfaces.action import Progress


class Action_Server(Node):

    def __init__(self):
        super().__init__('progress_action_server')
        # Create the action server
        self._action_server = ActionServer(
            self,
```

```
            Progress,
            'get_sum',
            self.execute_callback)
        self.get_logger().info('Action server started!')

    def execute_callback(self, goal_handle):
        self.get_logger().info('Starting to execute the task....')


        # Generate continuous feedback;
        feedback_msg = Progress.Feedback()

        sum = 0
        for i in range(1, goal_handle.request.num + 1):
            sum += i
            feedback_msg.progress = i / goal_handle.request.num
            self.get_logger().info('Continuous feedback: %.2f' % feedback_msg.progress)
            goal_handle.publish_feedback(feedback_msg)
            time.sleep(1)

        # Generate the final response.
        goal_handle.succeed()
        result = Progress.Result()
        result.sum = sum
        self.get_logger().info('Task completed!')

        return result

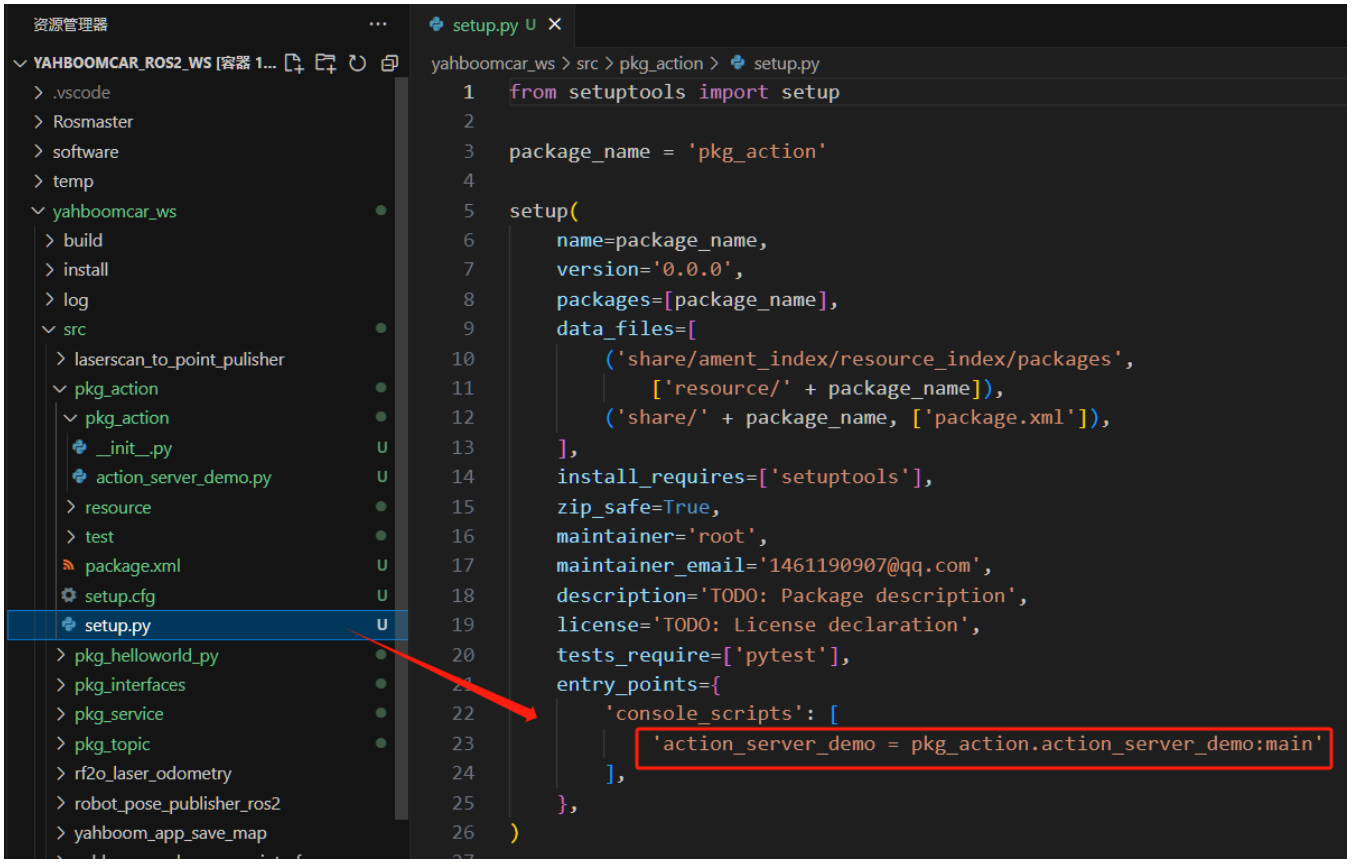

def main(args=None):

    rclpy.init(args=args)
    # Call the spin function and pass in the node object
    Progress_action_server = Action_Server()
    rclpy.spin(Progress_action_server)
    Progress_action_server.destroy_node()
    # Release resources
    rclpy.shutdown()
```

4.2 Edit the Configuration File

- Open setup.py and add the following to the console_scripts list

```
'action_server_demo = pkg_action.action_server_demo:main',
```



4.3 Build the Function Package

```
colcon build --packages-select pkg_action
```

4.4 Run the Program

```
ros2 run pkg_action action_server_demo
```

```
root@unbutu:~/yahboomcar_ros2_ws/yahboomcar_ws# ros2 run pkg_action action_server_demo
[INFO] [1698655793.746654229] [progress_action_server]: 动作服务已经启动!
```

Enter the following in another terminal:

```
ros2 action list
```

```
root@unbutu:~/yahboomcar_ros2_ws/yahboomcar_ws# ros2 action list
/get_sum
root@unbutu:~/yahboomcar_ros2_ws/yahboomcar_ws#
```

/get_sum is the action we need to call. Call it with the following command, enter in the terminal:

```
ros2 action send_goal /get_sum pkg_interfaces/action/Progress "{num: 10}"
```

Here we calculate the sum from 1 to 10:

```
root@unbutu:~/yahboomcar_ros2_ws/yahboomcar_ws# ros2 run pkg_action action_server_demo
[INFO] [1698656083.969110826] [progress_action_server]: 动作服务已经启动!
[INFO] [1698656086.785874392] [progress_action_server]: 开始执行任务....
[INFO] [1698656086.786558163] [progress_action_server]: 连续反馈: 0.10
[INFO] [1698656086.988069982] [progress_action_server]: 连续反馈: 0.20
[INFO] [1698656087.189707693] [progress_action_server]: 连续反馈: 0.30
[INFO] [1698656087.391339228] [progress_action_server]: 连续反馈: 0.40
[INFO] [1698656087.593014605] [progress_action_server]: 连续反馈: 0.50
[INFO] [1698656087.795406778] [progress_action_server]: 连续反馈: 0.60
[INFO] [1698656087.997475930] [progress_action_server]: 连续反馈: 0.70
[INFO] [1698656088.199655262] [progress_action_server]: 连续反馈: 0.80
[INFO] [1698656088.402166864] [progress_action_server]: 连续反馈: 0.90
[INFO] [1698656088.604606494] [progress_action_server]: 连续反馈: 1.00
[INFO] [1698656088.807212115] [progress_action_server]: 任务完成!
```

```
root@unbutu:~/yahboomcar_ros2_ws/yahboomcar_ws# ros2 action send_goal /get_sum pkg_interfaces/action/Progress num:\ 10\
Waiting for an action server to become available...
Sending goal:
     num: 10

Goal accepted with ID: f97430c5b72e464995b3b396c4f40226

Result:
    sum: 55

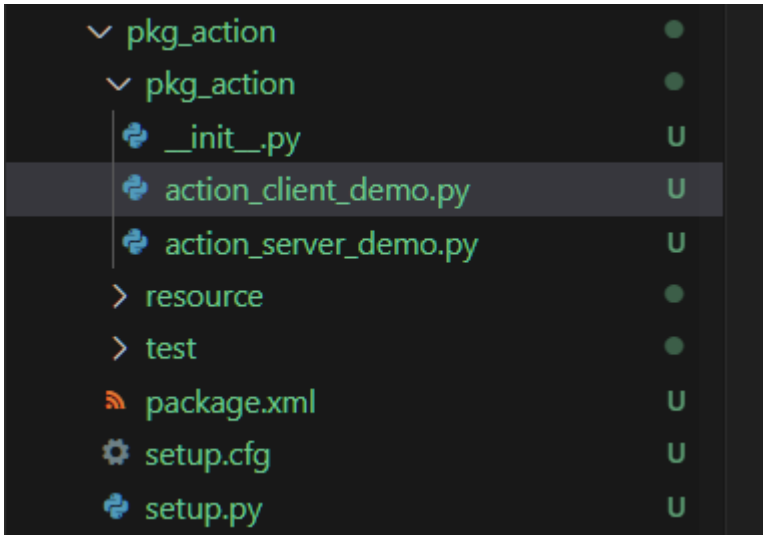
Goal finished with status: SUCCEEDED
root@unbutu:~/yahboomcar_ros2_ws/yahboomcar_ws#
```

The upper part of the figure above is the server, and the lower part is the client. You can see that during the calculation of the sum from 1 to 10, the server keeps feeding back the computation progress, and finally it shows that the task is completed. The client also receives the feedback that the sum is 55

5. Client Implementation

5.1 Create the Client

Create a new file [action_client_demo.py] in the same directory as [action_server_demo.py]



Next, edit [action_client_demo.py] to implement the server function and add the following code:

```python
import rclpy
from rclpy.action import ActionClient
from rclpy.node import Node
from pkg_interfaces.action import Progress

class Action_Client(Node):
```

```
    def __init__(self):
        super().__init__('progress_action_client')
        # Create the action client;
        self._action_client = ActionClient(self, Progress, 'get_sum')

    def send_goal(self, num):
        # Send the request;
        goal_msg = Progress.Goal()
        goal_msg.num = num
        self._action_client.wait_for_server()
        self._send_goal_future = self._action_client.send_goal_async(goal_msg,
feedback_callback=self.feedback_callback)
        self._send_goal_future.add_done_callback(self.goal_response_callback)

    def goal_response_callback(self, future):
        # Process the feedback after the goal is sent;
        goal_handle = future.result()
        if not goal_handle.accepted:
            self.get_logger().info('Request rejected')
            return

        self.get_logger().info('Request accepted, starting to execute the task!')

        self._get_result_future = goal_handle.get_result_async()
        self._get_result_future.add_done_callback(self.get_result_callback)

    # Process the final response.
    def get_result_callback(self, future):
        result = future.result().result
        self.get_logger().info('Final calculation result: sum = %d' % result.sum)
        # 5. Release resources.
        rclpy.shutdown()

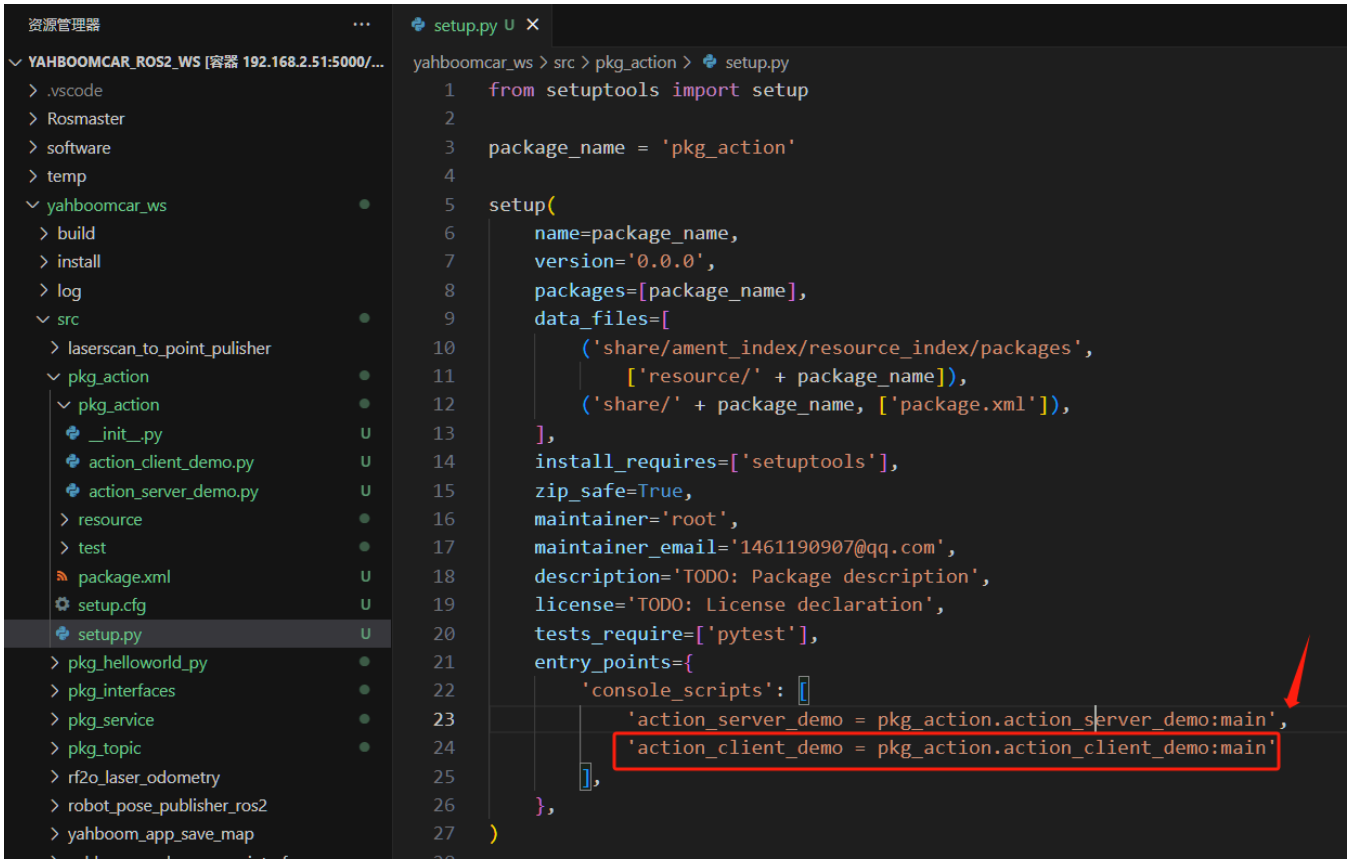
    # Process the continuous feedback;
    def feedback_callback(self, feedback_msg):
        feedback = (int)(feedback_msg.feedback.progress * 100)
        self.get_logger().info('Current progress: %d%%' % feedback)


def main(args=None):
    rclpy.init(args=args)
    action_client = Action_Client()
    action_client.send_goal(10)
    rclpy.spin(action_client)
```

5.2 Edit the Configuration File

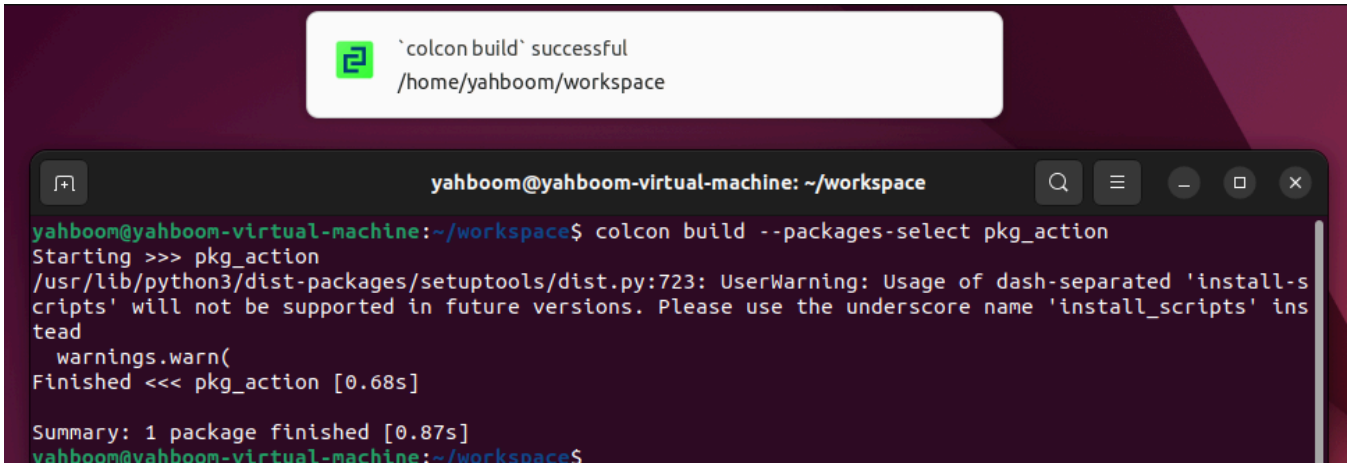
- Open setup.py and add the following to the console_scripts list

```
'action_client_demo = pkg_action.action_client_demo:main'
```



5.3 Build the Function Package

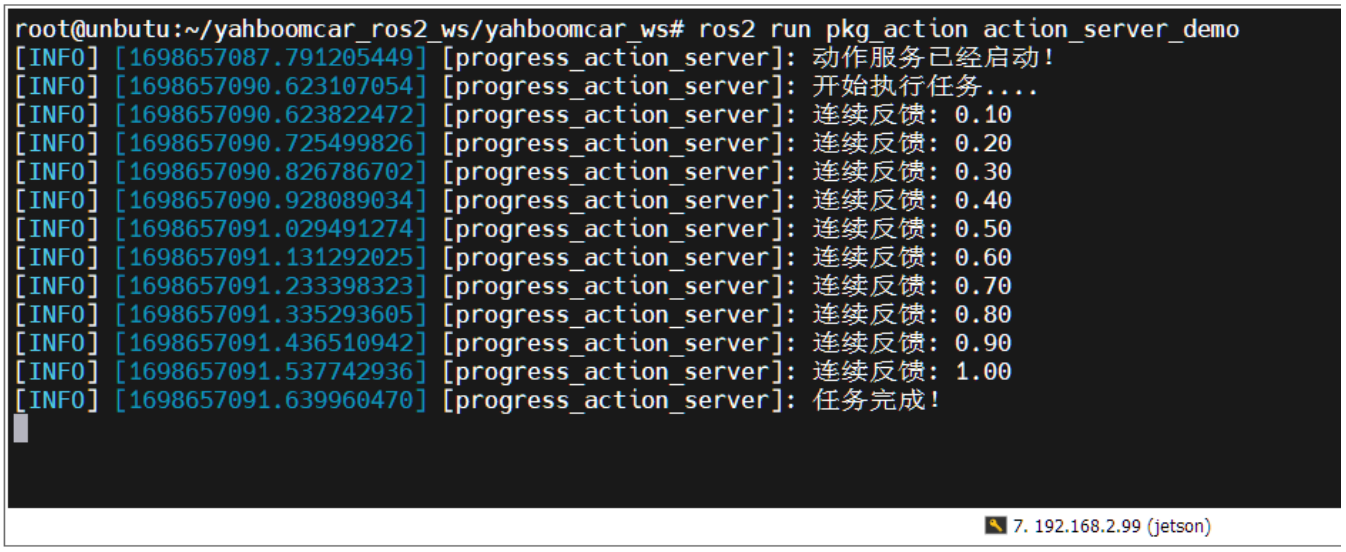
```
colcon build --packages-select pkg_action
```

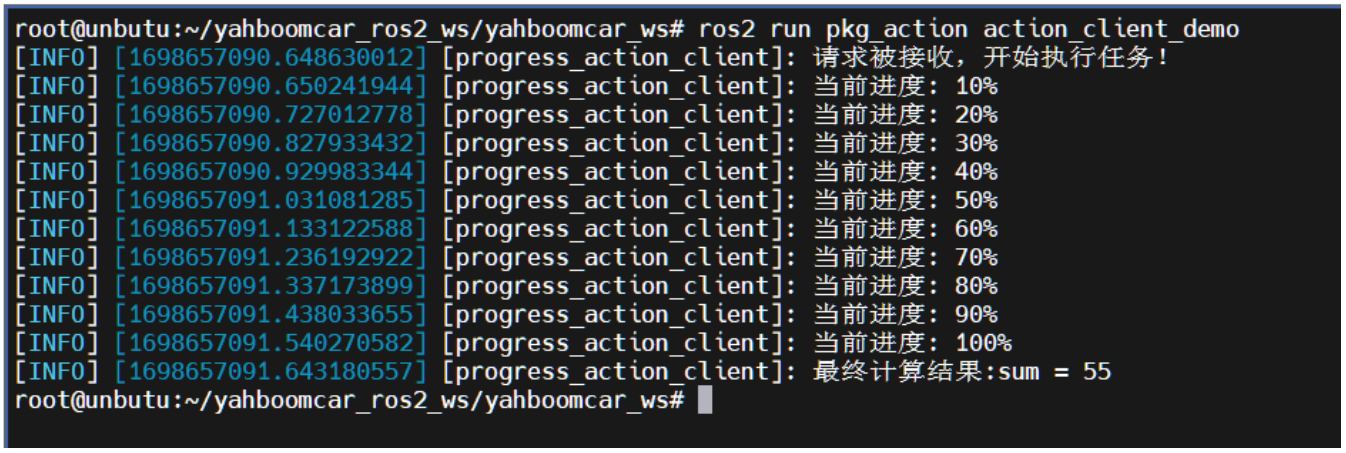


5.4 Run the Program

Execute the following in separate terminals:

```
#Start the server node
ros2 run pkg_action action_server_demo
#Start the client node
ros2 run pkg_action action_client_demo
```





The upper part of the figure above is the server, and the lower part is the client. Here we calculate the sum from 1 to 10, and you can see that during the calculation the server keeps feeding back the computation progress, and finally it shows that the task is completed. The client also receives the feedback that the sum is 55